ADVANCED NETWORKING TEAM PROJECT | APRIL 2025

AI-IDS: making detection approaches inspectable

Implemented and compared rule-based detection and a machine-learning workflow.

Rule-based path

Custom Snort rules targeted SSH connection bursts and MySQL-related payload patterns. Scapy-generated traffic, captures, and logs were used to inspect alerts.

Machine-learning path

An XGBoost classifier used CIC-IDS2017 flow data and packet-derived features. The team examined preprocessing, feature alignment, deployment, and debugging challenges.

My contribution

Traffic classification, packet-derived feature extraction, custom rules, and validation with generated traffic as part of the team comparison.

Evidence boundary

No comparable held-out detection metrics are asserted here. An SSH SYN-burst rule detects connection frequency, not proof of failed authentication.

Intrusion Detection System (IDS) with ML | Team report and demonstration, April 2025.